



Department of Computer Science and Engineering
Final Examination Summer 2024
CSE 321: Operating Systems

Duration: 1 Hour 40 Minutes

Total Marks: 45

Answer the following questions.
 Figures in the right margin indicate marks.

1. CO2	a) Find outputs of the given code: <pre> int main(){ pid_t p,q; int y[]={17,28}; p=fork(); if(p<0){ printf("fork failed\n"); } else if(p>0){ y[0]=y[0]*5; y[1]=y[1]/2; wait(); y[0]=y[0]+y[1]; y[1]=y[1]-y[0]; } else{ y[0]=y[0]+5; y[1]=y[1]-5; } q=fork(); if(q<0){ printf("fork failed\n"); } else if(q==0){ y[0]=y[0]%2; y[1]=y[1]%2; } else{ wait(); y[0]=y[0]*y[0]; y[1]=y[1]*y[1]; } for(int j=0;j<2;j++){ printf("y[%d] = %d\n",j,y[j]); } return 0; } </pre>	[4]
-----------	--	------------

[6]

b) Potato Bank Limited has two employees, Shoumo and Dipu who are responsible for maintaining the balance of a shared account. The system allows Shoumo and Dipu to either deposit or withdraw money from the account. To ensure the integrity of the account balance, the system needs a synchronization mechanism so that only one employee can modify the account balance at any time. Shoumo and Dipu perform **5 operations** each on the bank account having a starting balance of **2200 taka**. After performing operations on the account, both the employees get feedback from the customer which is equivalent to **3 operations**. The operations Shoumo and Dipu performs are given below:

Shoumo	Dipu
Deposit 1000	Withdraw 1000
Withdraw 2000	Withdraw 500
Deposit 5500	Deposit 3000
Withdraw 1100	Withdraw 500
Withdraw 1200	Deposit 300

Now **apply** Peterson's solution to simulate the above operations so that the integrity of the account balance is preserved. **Show** the account balance after each operation.

The constraints and initial condition of the system is below:

- Each operation takes **4s** to complete.
- Each employee must hand over the system access to the other employee every **12s**.
- Shoumo ($i=1, j=0$).
- Dipu ($i=0, j=1$).
- $turn = 1$.
- $flag[0] = true, flag[1] = true$.
- Dipu starts working first.

The algorithm for Peterson's solution is provided below:

```
do {
    flag[i] = true;
    turn = j;
    while( flag[j] && turn==j);
    //Critical Section
    flag[i] = false;
    // Remainder section
```

	<pre>} while(true);</pre>	
--	---------------------------	--

2. CO1	<table><tr><th>Processes</th><th>Arrival time</th><th>Burst time</th><th>Priority</th></tr><tr><td>P1</td><td>42</td><td>16</td><td>3</td></tr><tr><td>P2</td><td>62</td><td>13</td><td>1</td></tr><tr><td>P3</td><td>1</td><td>23</td><td>3</td></tr><tr><td>P4</td><td>59</td><td>21</td><td>4</td></tr><tr><td>P5</td><td>19</td><td>16</td><td>2</td></tr><tr><td>P6</td><td>61</td><td>20</td><td>1</td></tr></table>	Processes	Arrival time	Burst time	Priority	P1	42	16	3	P2	62	13	1	P3	1	23	3	P4	59	21	4	P5	19	16	2	P6	61	20	1	
Processes	Arrival time	Burst time	Priority																											
P1	42	16	3																											
P2	62	13	1																											
P3	1	23	3																											
P4	59	21	4																											
P5	19	16	2																											
P6	61	20	1																											
	Consider the following set of processes above, with the length of the CPU burst times given in milliseconds. Draw the Gantt charts illustrating the execution of these processes using preemptive priority (smaller number implies a higher priority) and Round Robin (time quantum = 9 ms) scheduling. Calculate average waiting time and average turnaround time for above scheduling algorithms and identify which algorithm is the best with proper justification.	[3+2 +3+2 +1]																												
3. CO3	a) Given memory partitions of 520k, 650k, 420k, 239k, 300k and 850k (in order, bottom to top). Apply first fit and best fit algorithms to place processes with space requirements of 417k, 610k, 415k, 510k and 390k . A constraint is being maintained which is that only one process can be allocated in a partition and in the main memory lower to higher memory addresses are arranged in top to bottom order. Explain which algorithm makes the most effective use of memory and find whether there is any external fragmentation or not.	[2+2 +2]																												
	b) A process of 16,063 Bytes needs four frames to fit in the memory. It causes an internal fragmentation of 321 Bytes . Find the size of each page in KB. [1 KB = 1024 Bytes]	[2]																												

	c) Consider a system with TLB miss ratio of 20%, and memory access time of 105 ns. If Effective Access Time is 144 ns, find TLB access time.	[2]																								
4. CO3	a) In a system the size of the main memory is 30 Bytes and size of each page is 2 Bytes. In a certain moment CPU generates logical addresses 3, 6, 15, 1 and 4 of a process respectively. Logical memory and page table of the process is given below. <table><tr><th colspan="2">Logical Memory</th></tr><tr><td>P₀</td><td>AB</td></tr><tr><td>P₁</td><td>CD</td></tr><tr><td>P₂</td><td>EF</td></tr><tr><td>P₃</td><td>GH</td></tr></table><table><tr><th colspan="2">PMT</th></tr><tr><td>P₀</td><td>13</td></tr><tr><td>P₁</td><td>19</td></tr><tr><td>P₂</td><td>7</td></tr><tr><td>P₃</td><td>12</td></tr></table><table><tr><th colspan="2">Main Memory</th></tr><tr><td colspan="2"></td></tr></table> i. Find corresponding physical addresses of generated logical addresses. If the physical address is not valid then write “invalid”. ii. Map user’s view of the main memory.	Logical Memory		P ₀	AB	P ₁	CD	P ₂	EF	P ₃	GH	PMT		P ₀	13	P ₁	19	P ₂	7	P ₃	12	Main Memory				[5+2]
Logical Memory																										
P ₀	AB																									
P ₁	CD																									
P ₂	EF																									
P ₃	GH																									
PMT																										
P ₀	13																									
P ₁	19																									
P ₂	7																									
P ₃	12																									
Main Memory																										
	b) Consider a computer with a main memory that has 3 frames and page reference string of 0-6 pages: [0 4 0 2 6 4 1 5 2 5 4 5]. The page reference string represents the order in which the pages are accessed by a program. Apply FIFO & LRU algorithm to simulate the page replacement that occurs when the main memory can hold at most 3 pages at a time and record the number of page faults. Now compare results and mention which algorithm performs better in this scenario.	[3+3+1]																								